Energy-Efficient 200 Gbit/s Transceivers for Optical Access Networks

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May 24, 2026

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Chapter 1

Access Networks and Coherent PON

1.1 Optical Access Networks

The access network is the final link between an operator's central exchange and the end user, often called the *last mile*. Exponentially growing bandwidth demands from streaming, cloud services, and 5G backhaul are placing severe pressure on this segment.

A passive optical network (PON) uses a point-to-multipoint (P2MP) topology in which a single optical line terminal (OLT) at the exchange serves multiple optical network units (ONUs) at the customer premises through a tree of passive optical splitters, as illustrated in fig. 1.1. No active components are required in the distribution plant, keeping operational costs low.

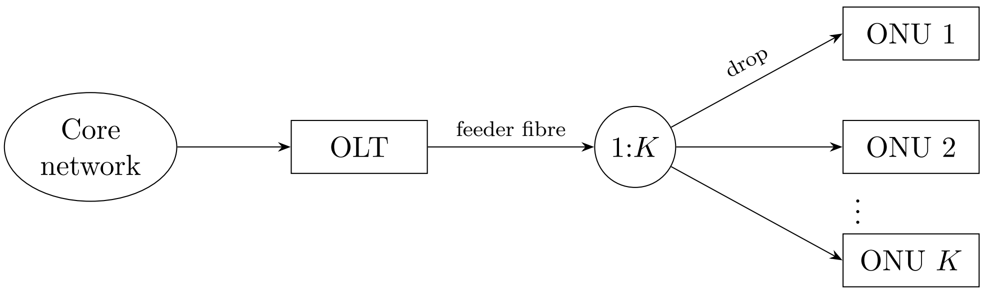


Figure 1.1: Passive optical network: one OLT serves K ONUs via a passive 1:K splitter.

1.2 Coherent PON

Traditional intensity-modulation direct-detection (IM-DD) passive optical networks are approaching their practical limits at data rates approaching 50 Gbps per wavelength. Coherent technology encodes information in the amplitude and phase of the optical signal, allowing modulation formats like QPSK that can carry more bits per symbol. Coherent receivers are also more sensitive, allowing for lower transmit powers. These factors make them a promising alternative to achieve rates of 200 Gbps on an acceptable power budget [6].

1.3 Specification

The Coherent Passive Optical Network (CPON) specification [1] defines a 100 Gbps single-wavelength P2MP system. This will be used as an example to test receiver implementations. This report focuses on the *downstream* direction (OLT to ONU) where receiver optimisations can be shared by all ONUs; the key physical-layer parameters are described in the following sections.

1.3.1 Modulation and Symbol Rate

The downstream signal uses non-differential DP-QPSK at a symbol rate of $R_s = 30.5\,\text{GBd}$. Each polarisation carries 2 bits per symbol by mapping consecutive bit pairs to in-phase and quadrature amplitudes $I, Q \in \{-1, +1\}$, giving QPSK constellation points at $\pm 1 \pm j$. Training and pilot symbols use the same mapping but at amplitude ± 3, placing them outside the data constellation for unambiguous identification.

1.3.2 Downstream Frame Structure

The fundamental framing unit is the *DSP subframe* of 3 712 symbols, organised as 116×32-symbol blocks. Every block begins with a pilot symbol; the first block also carries an 11-symbol training sequence, whose first symbol doubles as pilot index 1.

1.3.3 Key System Tolerances

The standard supports a 35 dB loss budget from OLT to ONU, which it states will allow a link length of 80 km with a 1:16 splitting ratio or 20 km with a 1:512 splitting ratio. To account for impairments in the core network, an SNR of 35 dB at the OLT must be supported. The OLT transmitter laser must be within $\pm 1.5\,\text{GHz}$ of its DWDM channel centre, and the ONU must acquire signals within $\pm 1.8\,\text{GHz}$ of the grid. In the worst case, the combined transmitter and LO offsets yield a carrier frequency offset (CFO) of up to $\approx 3\,\text{GHz}$ that the receiver DSP must correct. Both OLT and ONU lasers must have a linewidth of at most 1 MHz.

The specification requires a pre-FEC bit error ratio of $\leq 2 \times 10^{-2}$, which the soft-decision LDPC FEC reduces to a post-FEC BER of $\leq 10^{-12}$.

Chapter 2

Network and Receiver Models

2.1 Simulation Framework

All DSP algorithms in this report are designed, characterised, and compared through software simulation in MATLAB. The simulated receiver processes DP-QPSK symbols generated at the CPON downstream parameters described in chapter 1.

2.1.1 Fixed-Point Arithmetic

The receiver is evaluated with fixed-point arithmetic, as used in ASIC designs. This captures the effect of quantisation error on each implementation's performance, and underpins a bit-accurate energy model that goes beyond algorithmic complexity. All fixed-point modelling uses the MATLAB Fixed-Point Designer toolkit.

Quantisation Error With b fractional bits the quantisation step is $\Delta = 2^{-b}$; round-to-nearest produces an error e modelled as a zero-mean uniform random variable on $[-\Delta/2, \Delta/2]$ with variance

$$\sigma_q^2 = \frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} e^2 \,\mathrm{d}e = \frac{\Delta^2}{12} = \frac{2^{-2b}}{12}. \tag{2.1}$$

For a signal taking values over $[-A, A]$, $\Delta = \frac{2A}{2^b}$ and the signal-to-quantisation-noise ratio in dB with signal energy E_s is

$$\mathrm{SQNR} = 10\log_{10}\left(\frac{E_s}{\sigma_q^2}\right) = 6.02b + 10\log_{10}\left(\frac{3E_s}{A^2}\right). \tag{2.2}$$

CORDIC Coherent receivers perform phase estimates and rotations on complex-valued data in several DSP blocks. This is done efficiently in fixed-point hardware with the CORDIC algorithm, which evaluates these operations using only shifts and additions. CORDIC iteratively rotates a vector by the fixed angles $\arctan(2^{-i})$, each implemented as a bit-shift and an add/subtract. Each iteration adds roughly one bit of angular resolution, so n bits of precision require n CORDIC iterations and the cost is taken to be equivalent to an n-bit real multiplication.

2.1.2 System Parameters

The network under investigation assumes an end-to-end OLT-to-ONU loss budget of 35 dB and a back-to-back transmitter SNR of 35 dB, both taken from the CPON downstream specification of chapter 1. The two configurations are 80 km with a 1:16 splitting ratio and 20 km with a 1:512 splitting ratio. The fibre and signal parameters assumed for the network are listed in table 2.1.

Table 2.1: System parameters.

Parameter	Value
Central wavelength (λ)	1550 nm
Channel bandwidth (B)	30.5 GHz
Chromatic dispersion coefficient (D)	17 ps/(nm·km)
PMD coefficient (D_{DGD})	0.1 ps/$\sqrt{\text{km}}$

2.1.3 Pulse Shaping

To approach a signal bandwidth equal to the symbol rate R_s, the transmitter shapes each symbol with a root-raised-cosine (RRC) pulse and the receiver applies a matched RRC filter. The cascade has a raised-cosine spectrum, which is strictly band-limited to $(1+\beta)R_s/2$ on either side of the carrier and whose impulse response has zero crossings at every non-zero multiple of the symbol period, so the Nyquist no-ISI criterion is met at the sampling instants. The excess bandwidth is set by the roll-off factor $\beta \in [0,1]$; small β approaches the Nyquist minimum bandwidth of R_s but lengthens the pulse and raises its peak-to-average power.

2.1.4 Channel Impairments

Between the transmitter and receiver, the ideal symbol stream is degraded by the optical channel. Each impairment is applied as a separate stage so that individual DSP blocks can be characterised in isolation or in combination. The models used [12] are described below.

Additive White Gaussian Noise AWGN models the combined effect of shot noise and thermal noise. In simulation, the power of the incoming signal is measured and zero-mean Gaussian noise is added independently to each polarisation at a level set by the target SNR.

Chromatic Dispersion Chromatic dispersion (CD) arises because the group velocity of light in fibre depends on wavelength, so the frequency components of a pulse accumulate a frequency-dependent delay that spreads it over many symbol periods. It is an all-pass effect with a quadratic phase response. It is simulated by transforming the signal to the frequency domain with an FFT, applying the transfer function

$$H(\omega) = \exp\left(j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{2.3}$$

where D is the dispersion coefficient, λ the central wavelength, L the fibre length, and c the speed of light, then transforming back with an IFFT.

Polarisation-Mode Dispersion Polarisation-mode dispersion (PMD) results from random birefringence in the fibre, which causes the two polarisation modes to travel at slightly different speeds and couple into one another along the link. It is simulated as a concatenation of N_{pmd} birefringent sections, each applying a differential group delay of $\pm\tau/2$ to its two principal states in the frequency domain, with a random unitary rotation (the singular vectors of a complex Gaussian matrix) modelling the mode coupling between sections. The mean DGD scales as $\tau \propto D_{\text{PMD}}\sqrt{L}$, with D_{PMD} the fibre PMD coefficient.

Carrier Frequency Offset A carrier frequency offset (CFO) occurs because the transmitter and local-oscillator lasers are free-running and never exactly aligned in centre frequency, as discussed in section 1.3.3. The resulting beat appears as a linear phase ramp on the received samples. It is simulated by multiplying the signal by $\exp(j\,2\pi\,\Delta f\,T\,k)$, where Δf is the frequency offset, T the sample period, and k the sample index.

Phase Noise Phase noise originates from the finite linewidth of the transmitter and LO lasers, whose phases follow a random walk. It is modelled as a Wiener process: independent Gaussian per-symbol phase increments with variance $\sigma^2 = 2\pi\,\Delta\nu/R_s$, where $\Delta\nu$ is the combined laser linewidth, are accumulated and applied as a multiplicative phase $\exp(j\phi[k])$.

2.2 Energy Estimation

A DSP technique's energy efficiency cannot be judged from the receiver alone: a simpler implementation typically requires a higher SNR, which the transmitter must compensate for with extra launch power. The receiver contribution is modelled from fixed-point operation counts, and the transmitter contribution from the minimum SNR needed to meet the pre-FEC BER threshold in the CPON standard [1]. The two are combined to give a per-bit system energy.

2.2.1 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. The reference values shown in table 2.2 provide a useful estimate to compare DSP algorithms but a more accurate estimate would require power simulations for an ASIC implementation.

Table 2.2: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [7].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

Dependence on Word Length

A ripple-carry adder performs n bit-wise additions on words of length n, so its energy scales linearly with n. A shift-and-add multiplier accumulates n partial products of length n and so scales as n^2. Modern adders and multipliers (e.g. carry-lookahead, Booth/Wallace) reduce these costs by a constant factor or to $n \log n$, so the scalings used here are upper bounds; comparisons between algorithms remain valid because all are penalised equally. The proportionality constants are calibrated by a least-squares fit through the origin to the two data points of table 2.2, giving

$$E_A(n) = 3.16\, n \text{ fJ}, \qquad E_M(n) = 3.03\, n^2 \text{ fJ}. \tag{2.4}$$

Energy consumption for arithmetic operations in modern process nodes is not widely reported, and Dennard scaling (the rule that transistor power density remained constant as they were scaled down) [5] can no longer be applied to extrapolate from the values in table 2.2. As an approximation, the $\approx 75\%$ reduction in total power dissipation at constant frequency reported for an Intel core moving from 45 nm to 14 nm CMOS [16] is taken as a proxy for the per-operation energy scaling. This conflates contributions from leakage, clocks, caches and I/O with the arithmetic energy of interest, so the resulting numbers should be read as an estimate rather than a calibrated figure. Applying this reduction to the 45 nm fit gives

$$E_A(n) = 0.79\, n \text{ fJ}, \qquad E_M(n) = 0.76\, n^2 \text{ fJ}. \tag{2.5}$$

Amortisation

Some DSP algorithms perform computations that are reused over multiple symbols. For example, carrier frequency offset (CFO) varies much slower than the symbol rate so can be estimated less frequently. For a computation that requires N_M multiplications and N_A additions and is used by S symbols, the operations per symbol would be N_M/S and N_A/S.

Total Energy and Power

Many DSP blocks operate on complex-valued data; each complex multiplication is counted as 4 real multiplications and 2 real additions, and each complex addition as 2 real additions. The energy per symbol for the entire receiver is then

$$E_{\text{rx},s} = \sum_{k=1}^{L} 2\left[N_{A,k} E_A(n_k) + N_{M,k} E_M(n_k)\right], \tag{2.6}$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarisations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s. The receiver power is then

$$P_{\text{rx}} = R_s E_{\text{rx},s}, \tag{2.7}$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\text{rx}} = \frac{E_{\text{rx},s}}{\log_2(M)}, \tag{2.8}$$

where M is the modulation order.

2.2.2 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 1.1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share.

Minimum Required SNR

The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

Noise

The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power referred to the optical input of a balanced coherent receiver is $2h\nu B_{\text{eff}}$, where the factor of 2 captures the contributions from the signal and LO arms of the balanced detector summed in quadrature, and B_{eff} is the receiver electrical bandwidth, set equal to the symbol rate by a matched bandpass filter after the 1:K splitter in fig. 1.1. The power per polarisation received by each ONU is $P_{\text{tx}}/2\Gamma K$, where Γ is the loss from OLT to ONU. This gives a per-ONU NSR of

$$\text{NSR} = \frac{4\Gamma K h\nu B_{\text{eff}}}{P_{\text{tx}}} + \text{NSR}_0, \tag{2.9}$$

where $\text{NSR} = 1/\text{SNR}$ and NSR_0 is the back-to-back NSR at the OLT. Rearranging gives the transmitted power as

$$P_{\text{tx}} = \frac{4\Gamma K h\nu B_{\text{eff}}}{\text{NSR} - \text{NSR}_0}. \tag{2.10}$$

Assuming Nyquist pulse-shaping and matched bandpass filtering at the receiver ($B_{\text{eff}} = R_s$), the energy per bit transmitted is

$$E_{\text{tx}} = \frac{P_{\text{tx}}}{2KR_s \log_2(M)} = \frac{2\Gamma h\nu}{(\text{NSR} - \text{NSR}_0)\log_2(M)}. \tag{2.11}$$

2.2.3 Combined Energy

Before calculating a combined energy contribution, the transmitted optical energy in eq. (2.11) must be converted into an energy drawn from the electrical power supply by estimating the efficiencies of the transmitting laser and the modulator. The dual-polarisation modulator (fig. 2.1) consists of two nested-Mach-Zehnder IQ modulators sharing a common laser through a polarisation beam splitter, with their outputs recombined by a polarisation beam combiner. Each IQ modulator has the transfer function [12]

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \frac{1}{2}\cos\left[\frac{v_I(t)\pi}{2V_\pi}\right] + \frac{j}{2}\cos\left[\frac{v_Q(t)\pi}{2V_\pi}\right]. \tag{2.12}$$

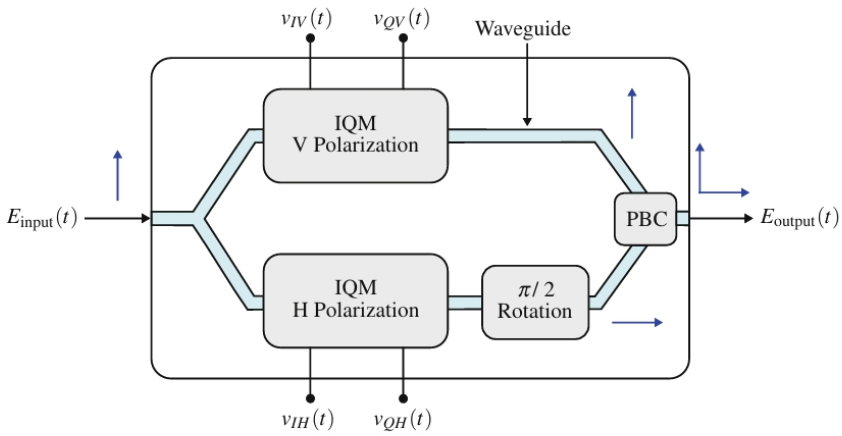


Figure 2.1: Dual-polarisation coherent modulator using MZMs.

With null biasing, QPSK is generated by driving each MZM between transmission minima, $v_I(t), v_Q(t) \in \{0, 2V_\pi\}$, so

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \pm\frac{1}{2} \pm \frac{j}{2} \tag{2.13}$$

and $P_{\text{out}} = P_{\text{in}}/2$, where the loss comes from power coupled to unused ports in the 2×2 couplers. An ideal polarisation beam splitter and combiner introduce no further loss, so the *marginal* efficiency of the dual-polarisation modulator is

$$\eta_{\text{DPM}} = \frac{dP_{\text{tx}}}{dP_{\text{in}}} = 0.5. \tag{2.14}$$

The overall efficiency would also include the power dissipated in the drive electronics, but this is independent of transmit power and so cancels in a comparison of receiver implementations. Combining with the marginal laser efficiency $\eta_{\text{laser}} = dP_{\text{opt}}/dP_{\text{elec}}$ gives a marginal wall-plug efficiency

$$\eta_{\text{WPE}} = \eta_{\text{laser}}\eta_{\text{DPM}} = 0.5\eta_{\text{laser}}. \tag{2.15}$$

η_{laser} will depend on the laser assembly and includes the increase in electrical power to supply the transmit laser and cooling apparatus. The combined energy contribution can now be estimated as

$$E_{\text{sys}} = \frac{1}{\eta_{\text{WPE}}}E_{\text{tx}} + E_{\text{rx}}. \tag{2.16}$$

This does not give the total energy consumed by the optical network, but a difference in E_{sys} shows the energy savings across receiver implementations.

Chapter 3

Equalisation

3.1 Introduction

The receiver compensates two distinct channel impairments with two distinct equalisers. Chromatic dispersion is static and accurately characterised by the fibre parameters, so it is removed by a fixed filter derived analytically from the channel model. Polarisation-mode dispersion varies on millisecond timescales and is unknown a priori, so it is tracked by an adaptive equaliser that also absorbs residual ISI from pulse shaping. This chapter derives both stages and quantifies their per-symbol computational cost.

3.2 Static Equalisation

3.2.1 Equalisation Procedure

Chromatic dispersion is a static, all-pass impairment; the receiver compensates it by applying the inverse of the channel transfer function. From the channel model of chapter 2, this is

$$H_{\mathrm{eq}}(\omega) = \exp\left(-j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{3.1}$$

whose inverse Fourier transform is a finite-energy chirp

$$h(t) = \sqrt{\frac{c}{jD\lambda^2 L}}\;\exp\left(j\,\frac{\pi c\,t^2}{D\lambda^2 L}\right). \tag{3.2}$$

The equaliser may be realised either as a time-domain FIR per polarisation, with taps obtained by sampling eq. (3.2) at the receiver period T and truncating to N_{CD} coefficients, or in the frequency domain. As the data stream is continuous, the frequency-domain filter is applied block-by-block by overlap-save: each block of N samples is transformed with an FFT, multiplied by the precomputed samples of H_{eq}, and inverse transformed. Discarding the first $N_{\mathrm{CD}} - 1$ samples of every block removes the circular-convolution wrap-around, leaving $N - N_{\mathrm{CD}} + 1$ valid output samples per block.

3.2.2 Filter Length

The equaliser must span the channel memory introduced by dispersion. With RRC pulse shaping the signal bandwidth is $\Delta f \approx R_s$, and the minimum length needed to capture the dispersive impulse response is

$$N_{\mathrm{DS}} = \left\lceil \frac{DL\Delta\lambda}{T} \right\rceil = \left\lceil \frac{2\pi|\beta_2|L\Delta f}{T} \right\rceil, \tag{3.3}$$

where T is the receiver sampling period. The length required to also remove inter-symbol interference (ISI) from the pulse shape is longer than eq. (3.3) [13]; an experimentally verified value for several pulse shapes is given by [8]

$$N_{\mathrm{CD}} = \left\lceil \frac{6.67|\beta_2|LR_s}{T} \right\rceil. \tag{3.4}$$

With an oversampling rate of 2, this gives filter lengths of 6 and 22 taps for the 20 km and 80 km networks, respectively.

3.2.3 Computational Complexity

A radix-2 FFT of size N decomposes into $\log_2 N$ stages of $N/2$ butterflies. Each butterfly costs 4 real multiplications and 6 real additions, giving

$$N_M^{\mathrm{FFT}} = 2N\log_2 N, \qquad N_A^{\mathrm{FFT}} = 3N\log_2 N. \tag{3.5}$$

Each overlap-save block needs a forward FFT, an inverse FFT of equal cost, and N complex multiplications by the precomputed filter spectrum (the filter FFT is computed once offline because the channel is static). This is $2N_M^{\mathrm{FFT}} + 4N$ real multiplications and $2N_A^{\mathrm{FFT}} + 2N$ real additions per block, shared over the $N - N_{\mathrm{CD}} + 1$ valid outputs. Scaling by the oversampling ratio $S = f/R_s$ to express the cost per symbol gives

$$\begin{aligned} N_{M,\mathrm{OS}} &= S\,\frac{4N\log_2 N + 4N}{N - N_{\mathrm{CD}} + 1}, \\ N_{A,\mathrm{OS}} &= S\,\frac{6N\log_2 N + 2N}{N - N_{\mathrm{CD}} + 1}. \end{aligned} \tag{3.6}$$

In the time domain, direct convolution with the N_{CD}-tap complex filter costs N_{CD} complex multiplications and $N_{\mathrm{CD}} - 1$ complex additions per output sample. Converting to real operations and scaling by S, the per-symbol cost is

$$N_{M,\mathrm{TD}} = 4SN_{\mathrm{CD}}, \qquad N_{A,\mathrm{TD}} = S\left(4N_{\mathrm{CD}} - 2\right). \tag{3.7}$$

The FFT can be simplified further by rounding each twiddle factor to the nearest signed power of two in its real and imaginary parts, replacing every twiddle multiplication with a pair of bit-shifts and eliminating the real multiplications inside the FFT and IFFT entirely. As shown by the rightmost column of table 3.2, this reduces the multiplication count by roughly an order of magnitude at no additional cost in additions.

Operation	Time-domain	Overlap-save	Overlap-save (2^k twiddle)
Real multiplications	$4SN_{\mathrm{CD}}$	$S\dfrac{4N\log_2 N + 4N}{N - N_{\mathrm{CD}} + 1}$	$S\dfrac{4N}{N - N_{\mathrm{CD}} + 1}$
Real additions	$S\left(4N_{\mathrm{CD}} - 2\right)$	$S\dfrac{6N\log_2 N + 2N}{N - N_{\mathrm{CD}} + 1}$	$S\dfrac{6N\log_2 N + 2N}{N - N_{\mathrm{CD}} + 1}$

Table 3.1: Per-symbol computational cost of the time-domain and overlap-save chromatic dispersion equalisers, including overlap-save with twiddle factors rounded to a power of 2.

Evaluating these expressions for the two networks with 2 samples per symbol gives the per-symbol operation counts in table 3.2, where the FFT size N is chosen as the power of 2 that minimises the overlap-save multiplication count.

			Time-domain		Overlap-save		Overlap-save (2^k twiddle)	
L (km)	N_{CD}	N	RM	RA	RM	RA	RM	RA
20	6	32	48	44	56.9	75.9	9.48	75.9
80	22	128	176	172	76.6	105.3	9.57	105.3

Table 3.2: Filter length N_{CD}, optimum FFT size N, and real multiplications (RM) and real additions (RA) per symbol for each chromatic dispersion equaliser implementation on the 20 km and 80 km networks.

3.3 Adaptive Equalisation

3.3.1 Equalisation Procedure

PMD is tracked with a 2×2 MIMO equaliser comprising four finite impulse response (FIR) filters whose outputs are

$$x_{\mathrm{out}} = \mathbf{h}_{xx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{xy}^H \mathbf{y}_{\mathrm{in}}, \quad y_{\mathrm{out}} = \mathbf{h}_{yx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{yy}^H \mathbf{y}_{\mathrm{in}}. \tag{3.8}$$

The error terms are produced blindly using the constant modulus algorithm (CMA) [15], which exploits the fact that an ideal QPSK constellation lies on a circle of fixed radius. CMA minimises the cost function $J = \mathbb{E}\left[\left(|x_{\mathrm{out}}|^2 - R\right)^2\right]$, where the dispersion constant $R = \mathbb{E}[|s|^4]/\mathbb{E}[|s|^2]$ is set by the transmitted symbols s and reduces to a constant for QPSK. Taking the stochastic gradient of J and absorbing the factor of -2 into the step size yields the error estimates

$$e_x = \left(R - |x_{\mathrm{out}}|^2\right), \quad e_y = \left(R - |y_{\mathrm{out}}|^2\right). \tag{3.9}$$

The filter weights are then updated as

$$\begin{aligned} \mathbf{h}_{xx} &\leftarrow \mathbf{h}_{xx} + \mu e_x \mathbf{x}_{\mathrm{in}} x_{\mathrm{out}}^*, \quad \mathbf{h}_{xy} \leftarrow \mathbf{h}_{xy} + \mu e_x \mathbf{y}_{\mathrm{in}} x_{\mathrm{out}}^*, \\ \mathbf{h}_{yx} &\leftarrow \mathbf{h}_{yx} + \mu e_y \mathbf{x}_{\mathrm{in}} y_{\mathrm{out}}^*, \quad \mathbf{h}_{yy} \leftarrow \mathbf{h}_{yy} + \mu e_y \mathbf{y}_{\mathrm{in}} y_{\mathrm{out}}^*, \end{aligned} \tag{3.10}$$

where the convergence parameter μ is chosen to be a power of 2 for hardware simplicity. CMA suffers from the well-known singularity problem, in which both outputs converge to the same input signal. To avoid this, $\mathbf{h}_{xx}$ and $\mathbf{h}_{xy}$ are initialised with a single one at the centre tap; then, once x_{out} has converged, the weights producing y_{out} are initialised as $\mathbf{h}_{yx}[n] = \mathbf{h}_{xx}^*[-n]$ and $\mathbf{h}_{yy}[n] = -\mathbf{h}_{xy}^*[-n]$ [10].

Because the output terms are complex, the standard sign function generalises to a complex signum

$$\mathrm{csgn}(z) = \mathrm{sgn}(\mathrm{Re}\, z) + j\,\mathrm{sgn}(\mathrm{Im}\, z). \tag{3.11}$$

Retaining only $\mathrm{sgn}(e)$ and $\mathrm{csgn}(x_{\mathrm{out}})$ in eq. (3.10) eliminates every multiplication from the weight update at no performance penalty [2].

3.3.2 Filter Length

The filter must span the channel memory introduced by PMD. The mean differential group delay (DGD) is $\langle\Delta\tau\rangle = D\sqrt{L}$, where D is the DGD parameter and L is the link length. The instantaneous DGD is modelled by a Maxwellian distribution [11]

$$\mathbb{P}[\Delta\tau > x] \approx \frac{4}{\pi}\frac{x}{\langle\Delta\tau\rangle}\exp\left(-\frac{4x^2}{\pi\langle\Delta\tau\rangle^2}\right). \tag{3.12}$$

For a given outage probability, the corresponding tap count is

$$N = \left\lceil \frac{x}{T} \right\rceil, \tag{3.13}$$

where T is the receiver sampling period. The relevant values for the access network are summarised in table 3.3: the resulting DGD budget is covered by a single tap. However, with only one tap, the adaptive equaliser cannot also function as a matched filter [8], leaving residual ISI from pulse shaping.

Quantity	Value
Link length L	$\leq 80\,\text{km}$
DGD parameter D	$0.1\,\text{ps}/\sqrt{\text{km}}$
Mean DGD $\langle\Delta\tau\rangle$	$\leq 0.9\,\text{ps}$
Outage target	10^{-12}
Required span $x = 4.81\langle\Delta\tau\rangle$	$4.3\,\text{ps}$
Sampling period T ($S = 2$)	$16.4\,\text{ps}$
Required taps N	1

Table 3.3: PMD budget for the 80 km access network.

3.3.3 Computational Complexity

The per-symbol cost splits into three stages: forming the filter outputs, forming the error terms, and updating the weights. Forming the outputs evaluates the four N-tap FIR filters of eq. (3.8), and the four N-tap weight vectors are updated according to eq. (3.10). The sign-sign variant retains only $\operatorname{csgn}(e)$ and $\operatorname{csgn}(x_{\text{out}})$ [2], eliminating every multiplication in the update at no performance penalty. To support the 30.5 GBd symbol rate the equaliser is parallelised across P lanes, the input vector of lane p being

$$\mathbf{x}_p = [\mathbf{x}[mP + p], \mathbf{x}[mP + p - 1], \ldots, \mathbf{x}[mP + p - (N-1)]]\,. \tag{3.14}$$

The outputs are formed independently on every lane, costing $16NP$ real multiplications and $(8N + 4)P$ real additions per block of P symbols regardless of how the weights are updated. The two strategies for forming the errors and updating the weights differ in how they scale with P; in each case the per-symbol cost is obtained by dividing the block cost by $2P$.

CMA The CMA error terms of eq. (3.9) take the squared modulus of each output and subtract it from the dispersion constant R, formed on every lane. A single set of weights is shared across the lanes and updated from their summed contributions,

$$\Delta\mathbf{h} = \mu \sum_{p=0}^{P-1} e[p]\mathbf{x}_p x^*_{\text{out}}[p], \tag{3.15}$$

so both the error terms and the update scale with P, as summarised in table 3.4.

Stage	Real mult.	Real add.
Error terms	$4P$	$2P$
Weight update (direct)	$16NP + 4P$	$16NP$
Weight update (sign-sign)	0	$16NP$

Table 3.4: Cost of the CMA error terms and weight update per block, parallelised across P lanes.

Pilots Alternatively the update can be driven directly from the pilot symbol using a least mean-squares (LMS) approach [14]

$$\Delta\mathbf{h} = \mu e^* \mathbf{x}_p = \mu(\hat{s} - s)^* \mathbf{x}_p, \tag{3.16}$$

where s and $\hat{s}$ are the transmitted and estimated pilot symbols at the start of each block, and $\mathbf{x}_p$ is the buffer of N symbols used to form the pilot estimate from eq. (3.8). A sign-sign variant analogous to the CMA case is obtained by retaining only $\operatorname{csgn}(e)$. With only one pilot symbol per block the error is formed once and the weights updated once, so neither scales with P, as summarised in table 3.5.

Stage	Real mult.	Real add.
Error terms	0	4
Weight update (direct)	$16N$	$16N$
Weight update (sign-sign)	0	$16N$

Table 3.5: Cost of the pilot error terms and weight update per block, formed once per block.

3.4 Methodology

-rrc pulse shaping -monte carlo -application of CD and PMD - convergence parameter size and number of symbols before single spike and number of symbols discarded

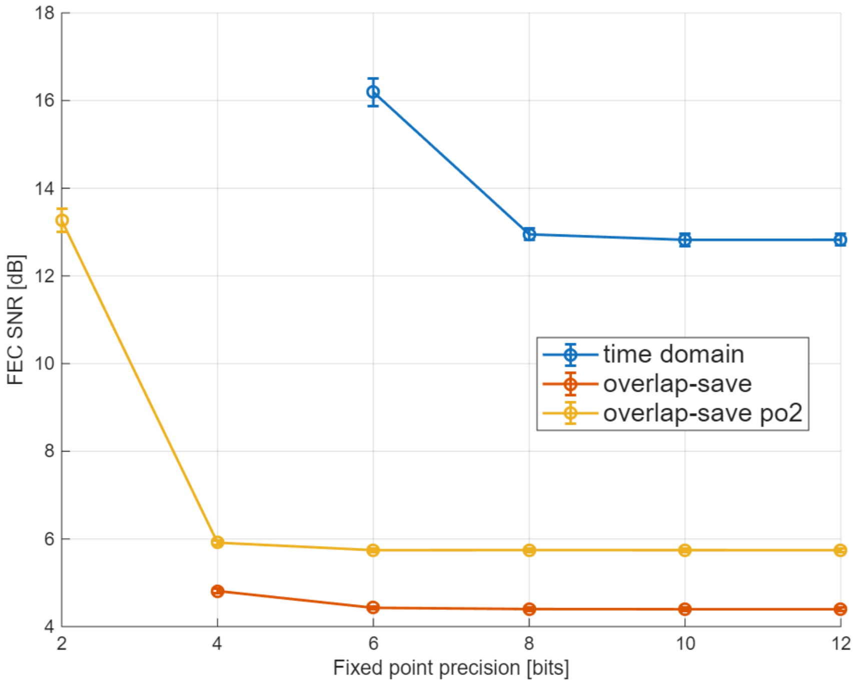
3.5 Results & Discussion

3.5.1 Static Equalisation

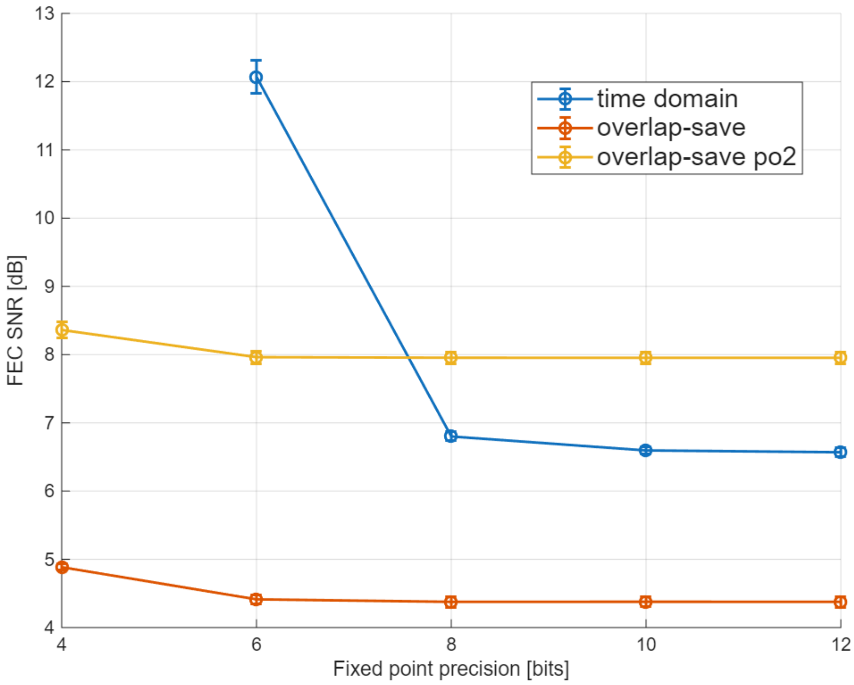
Figures 3.1a and 3.1b compare the performance of each CD equaliser implementation for the 20 km and 80 km networks. The time domain equaliser performs far worse on the 20 km network with 6 taps than on the 80 km network and exhibits a sharp increase in FEC SNR below 8 bits of precision. This suggests the time domain FIR is highly sensitive to quantisation noise and coarse taps. Overlap-save has similar performance on both networks and shows little change in FEC SNR down to 4 bits of precision but was not able to achieve the FEC limit at 2 bits of precision. Rounding twiddle factors to a power of 2 had less impact on performance for the 20 km network but was more costly for 80km. The energy and FEC SNR at the lowest precision before performance drops for each equaliser implementation are summarised in table 3.6. The time domain equaliser is not viable.

Network	Implementation	FL (bits)	Energy (fJ/bit)	FEC SNR (dB)
	Time-domain	8	2605	12.95 ± 0.13
20 km	Overlap-save	4	929	4.81 ± 0.05
	Overlap-save (2^k twiddle)	4	355	5.91 ± 0.05
	Time-domain	8	9181	6.80 ± 0.07
80 km	Overlap-save	4	1249	4.89 ± 0.05
	Overlap-save (2^k twiddle)	4	444	8.36 ± 0.12

Table 3.6: Energy per bit and FEC SNR (mean $\pm$ std over trials) for the minimum fixed-point precision that meets the FEC threshold: time-domain at $\mathrm{FL} = 8$ and overlap-save at $\mathrm{FL} = 4$.



(a) 20 km



(b) 80 km

Figure 3.1: FEC SNR of each CD equaliser implementation as fixed point precision varies for the 20 km and 80 km networks.

3.5.2 Adaptive Equalisation

Filter Length

Equation (3.13) predicts an adaptive filter length of 1 tap for both the 20 km and 80 km networks. Figure 3.2 indicates that increasing the tap length degrades performance such that the FEC threshold is not achieved. Adding more taps increases the dimension of the gradient descent in eq. (3.10) which can cause slower convergence to the minimum error, so this loss in performance is likely a failure to converge before the single spike reinitialisation.

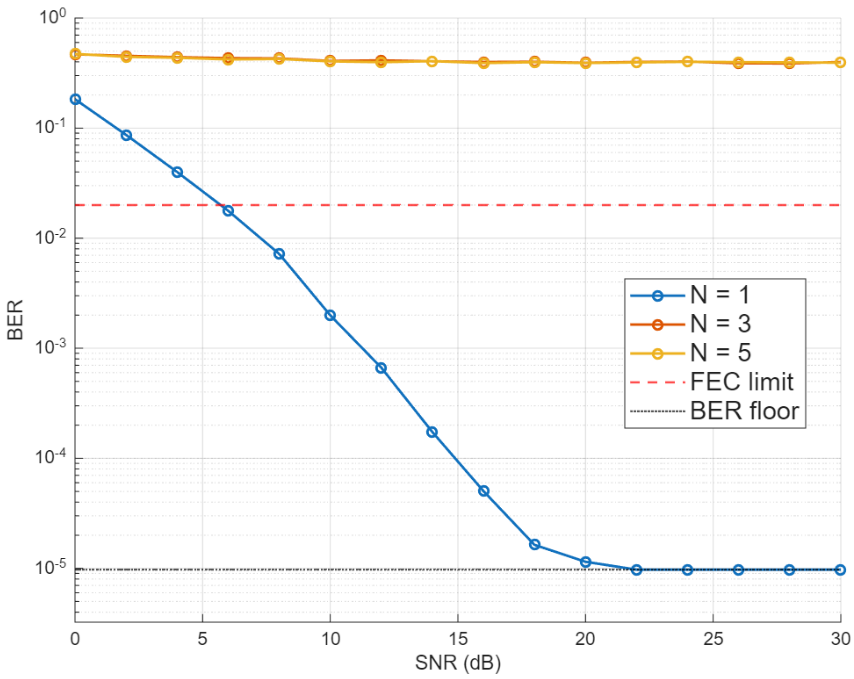


Figure 3.2: BER vs SNR for adaptive equaliser tap lengths of 1,3,5 using CMA and direct update at 16 bits of precision.

Algorithm Comparison

Table 3.7 totals the per-symbol real additions and multiplications of the output formation, error and weight-update stages for $N = 1$ tap and $P = 32$ lanes, the operating point used in the fixed-point sweep. The pilot variants are roughly half the multiplications and a quarter of the additions of CMA, since their error and update stages do not scale with P. The sign-sign variant of CMA halves the multiplication count without changing the additions, while the pilot sign-sign variant saves only the $16N = 16$ multiplications of the update.

Combination	Real mult.	Real add.
CMA, direct	20	15
CMA, sign-sign	10	15
Pilot, direct	8.25	6.3125
Pilot, sign-sign	8	6.3125

Table 3.7: Total real additions and multiplications per symbol for each algorithm at $N = 1$ tap and $P = 32$ lanes, including output formation, error terms and weight update.

Figure 3.3 shows the corresponding energy per bit for each combination as fixed point precision varies. Direct CMA uses $\approx 2\times$ the energy of sign-sign CMA, which in turn uses $\approx 1.5\times$ the energy of the pilot-based updates when fixed-point precision is equal.

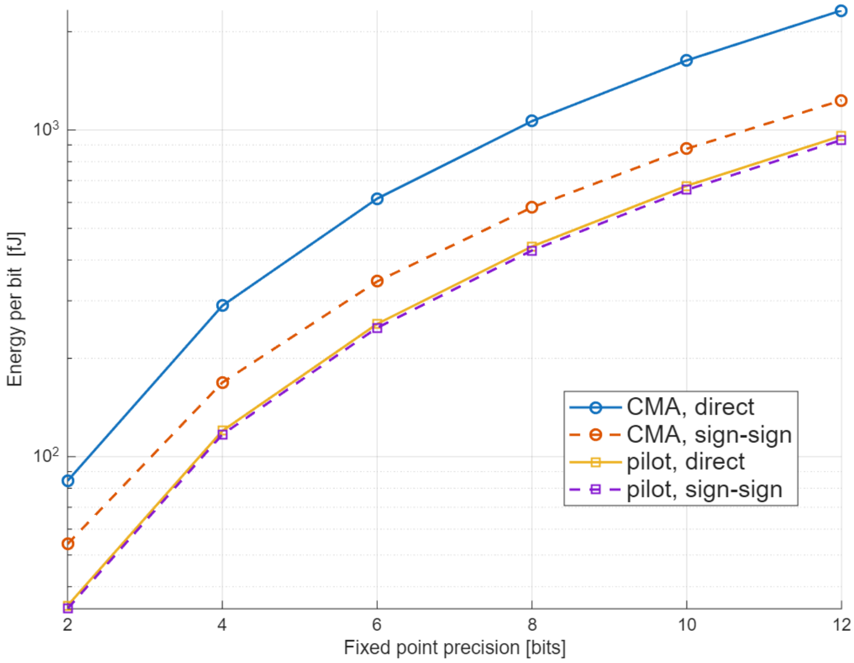
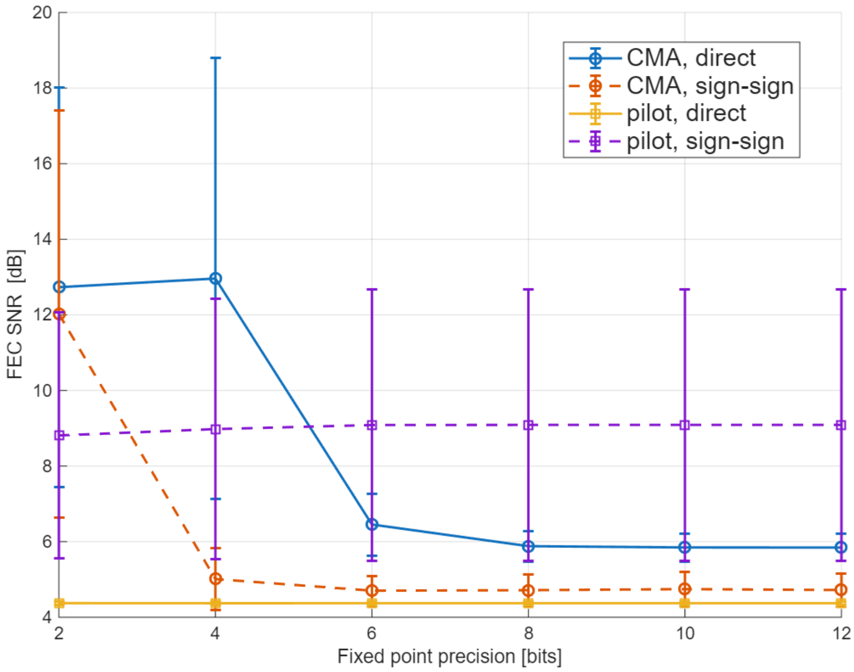


Figure 3.3: Energy per bit as fixed point precision of the input, output, and error terms vary for each adaptive equalisation implementation.

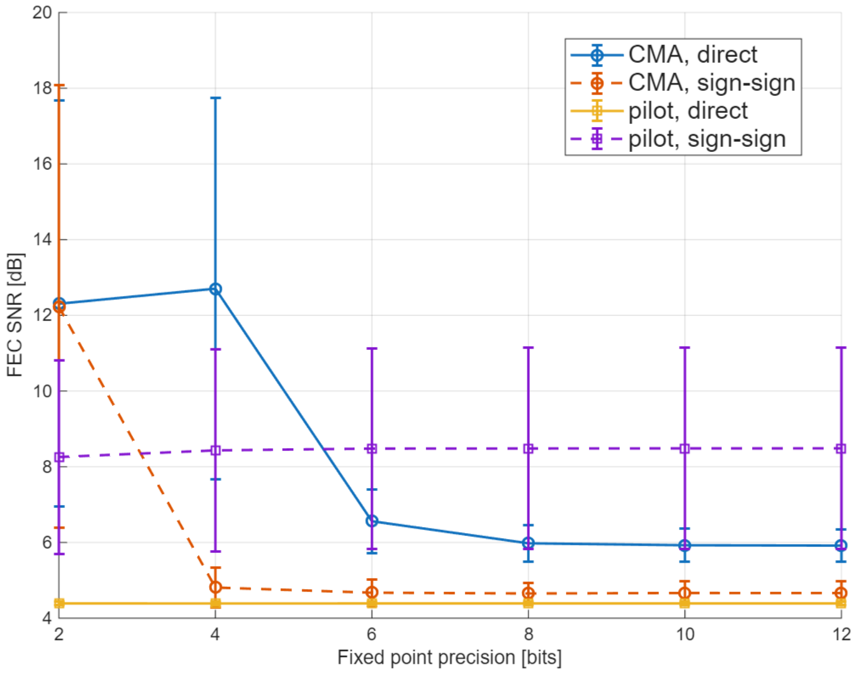
The FEC SNR of each implementation is shown in figs. 3.4a and 3.4b.The direct, pilot-based update shows the best performance at all precisions and works for a gradient estimate with only 2 bits of precision. Only using the sign of the error with pilots significantly degrades performance with negligible energy savings so is not viable for adaptive equalisation. Sign-sign CMA exhibits better performance than the direct version and can operate at 4 bits of precision without a significant increase in FEC SNR. Thus the two most promising implementations are direct pilot-based updates with 2 bits of precision or sign-sign CMA with 4 bits of precision which are compared in table 3.8. Direct, pilot-based updates appear to outperform sign-sign CMA in energy consumption and FEC SNR but this ignores the effects of CFO and phase error on equaliser performance. CMA is inherently robust to these effects as it computes the error from the modulus of the estimated symbol while the direct LMS estimate from the pilots can become very inaccurate. To realise the potential benefits of the pilot-based updates, there must be feedback from the carrier recovery section to allow the equaliser to operate on phase-corrected symbols. This is investigated in chapter 5.

Combination	Precision / bits	Energy (fJ/bit)	FEC SNR (dB)
Pilot, direct	2	34.97	4.37 ± 0.05
CMA, sign-sign	4	168.60	5.01 ± 0.82

Table 3.8: Energy per bit and FEC SNR (mean $\pm$ std over trials) of the two most promising adaptive equalisation operating points on the 20 km network at $N = 1$ tap.



(a) 20 km network



(b) 80 km network

Figure 3.4: FEC SNR vs fixed-point precision of the input, output, and error terms for each adaptive equalisation implementation on the 20 km and 80 km networks.

3.5.3 Combined Performance

- results for combined equalisation, FEC SNR vs fxp.

Chapter 4

Carrier Recovery

4.1 Introduction

A coherent receiver recovers the transmitted symbols by mixing the incoming optical signal with a local oscillator (LO) laser, allowing the amplitude and phase of each symbol to be extracted directly. However, the transmitter and LO lasers will in general have different centre frequencies and independent phase trajectories, producing a carrier frequency offset (CFO) and accumulated phase noise on the received symbols. Carrier recovery is the DSP stage responsible for estimating and removing these impairments before symbol decisions are made.

Carrier recovery is divided into two sequential stages. Frequency recovery performs a coarse estimate of the CFO and corrects it so that only slow residual phase variations remain. Phase recovery then tracks and removes these residual phase errors on a block-by-block basis. The algorithm chosen for each stage determines both the minimum achievable SNR and the receiver DSP energy, making algorithm selection central to the system-level energy comparison of chapter 2.

This chapter describes the Rife & Boorstyn (R&B) and differential phase and Kay frequency recovery algorithms, and the Viterbi & Viterbi (V&V) and pilots-only phase recovery algorithms, analysing their estimation accuracy and computational complexity. Suitable candidates for an energy-efficient receiver are identified and the criteria for selecting the most energy-efficient implementation are outlined.

4.2 Frequency Recovery

4.2.1 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4}+m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i+\theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i+4\theta[i])} + n'[i] \tag{4.1}$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise, which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i + \theta[i])} + n[i], \tag{4.2}$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$, which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4 \arg(x[i])\}|_{-\pi}^{\pi} \tag{4.3}$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \tag{4.4}$$

for data-aided operation, where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

4.2.2 Rife & Boorstyn

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \ldots, z[i+L]$ where L is the observation length. The Rife & Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give [9]

$$\delta = -\Re\left[\frac{X_{k+1} - X_{k-1}}{2X_k - X_{k-1} - X_{k+1}}\right]. \tag{4.5}$$

The updated index $k' = k + \delta$ is then scaled by the sampling rate to find the true CFO. The resolution in k is $\epsilon(k) = 1/2N_{FFT}$. The CPON specification provides only $L = 11$ training symbols, so data-aided operation without zero-padding gives $\epsilon(\hat{f}_d) \approx 0.05\,R_s$ — too coarse for the downstream phase recovery to correct. Zero-padding the training sequence to a larger N_{FFT} improves the resolution without requiring additional training data. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 4.1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarisations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) provides a lower bound on the achievable MSE and is given by [4]

$$\text{MCRB} = \frac{3}{2\pi^2 L^3 \times \text{SNR}}. \tag{4.6}$$

Rife & Boorstyn achieves this bound above a threshold SNR of ≈ 4 dB, where the threshold SNR is defined as the SNR below which the estimator begins to deviate from the MCRB. Below this, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

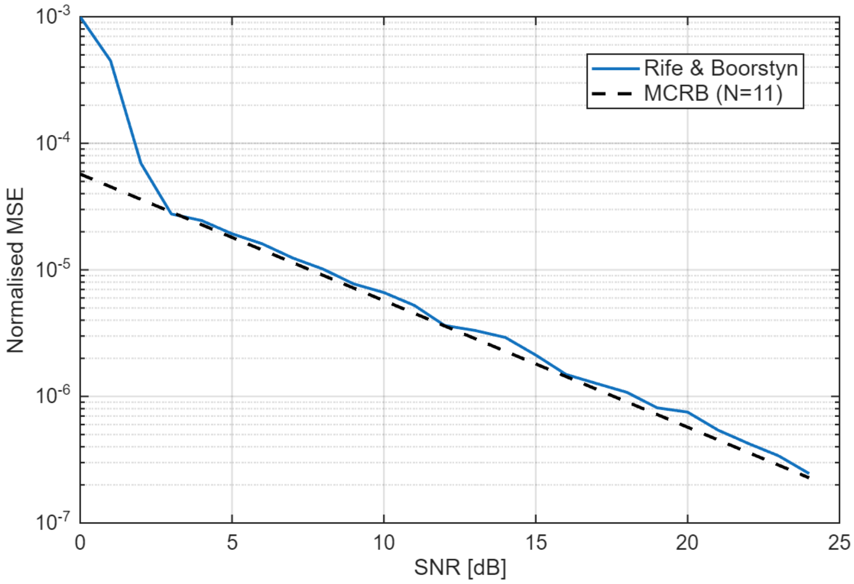


Figure 4.1: Normalised MSE of the Rife & Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the Rife & Boorstyn algorithm improves with longer observation lengths, as shown in fig. 4.2 for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

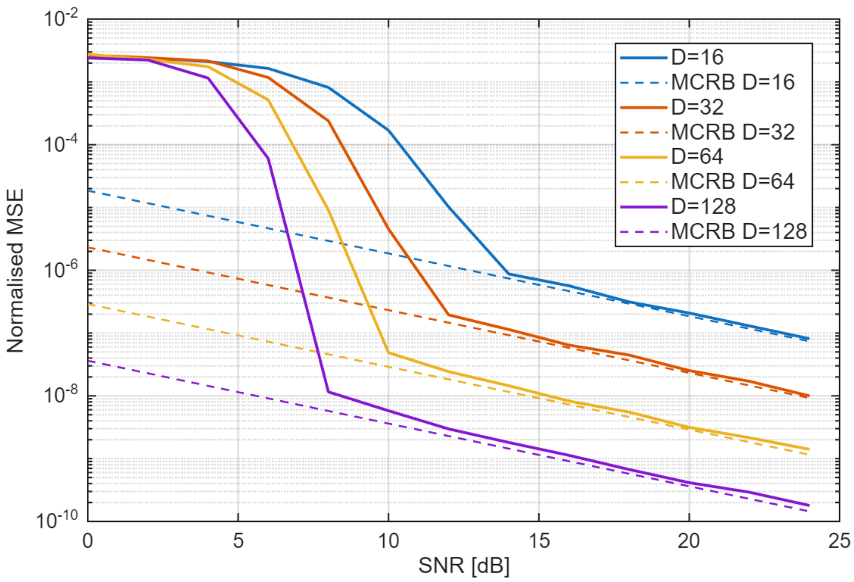


Figure 4.2: Normalised MSE of the Rife & Boorstyn algorithm for non-data-aided operation and different observation lengths.

4.2.3 Differential Phase and Kay Estimator

The Rife & Boorstyn algorithm exhibits good performance even at low SNRs, but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2}\log_2(N_{FFT})$ complex multiplications and $N_{FFT}\log_2(N_{FFT})$ complex additions. The fine search from eq. (4.5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \tag{4.7}$$

over several symbols and averaging to remove the effect of θ. Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$, so exhibits poor accuracy. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi}\sum_{k=1}^{L}\frac{6k(L-k)}{L(L^2-1)}\arg(z[k]z^*[k-1]). \tag{4.8}$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (4.7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB, but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. When the total phase drift $\Delta\phi = 2\pi f_d L T_s$ across the observation window exceeds 2π, a wrap shifts the estimated CFO by $\pm 1/(LT_s)$. A first-pass coarse CFO correction must therefore be applied before the Kay estimator to keep the residual CFO within the unambiguous range. The performance of the differential phase and Kay estimator for data-aided operation is shown in fig. 4.3. The MCRB is achieved above a threshold SNR of ≈ 7 dB. This is higher than for Rife & Boorstyn, but at a reduced computational cost. Unlike the Rife & Boorstyn algorithm, the threshold SNR of the differential phase and Kay estimator does not depend on observation length, as shown in fig. 4.4.

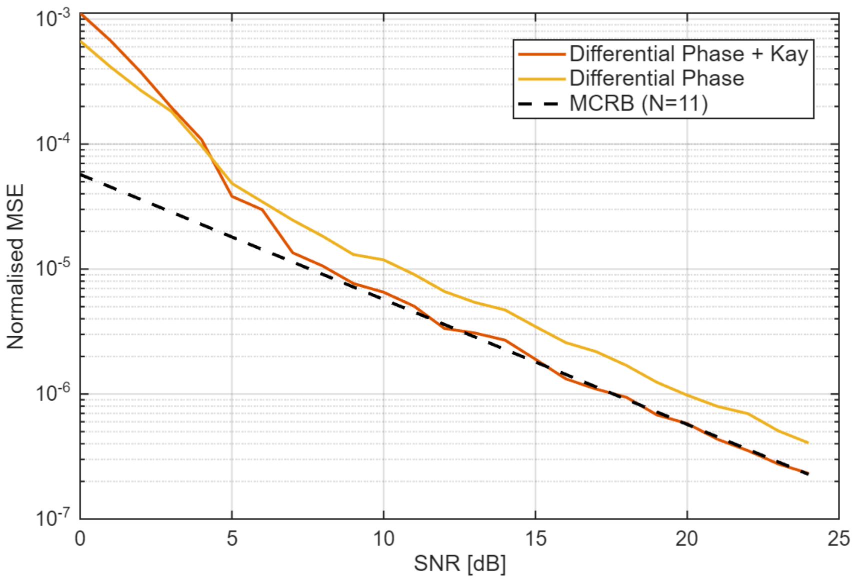


Figure 4.3: Normalised MSE of frequency estimation with the differential phase and Kay estimator for $L = 11$ training symbols.

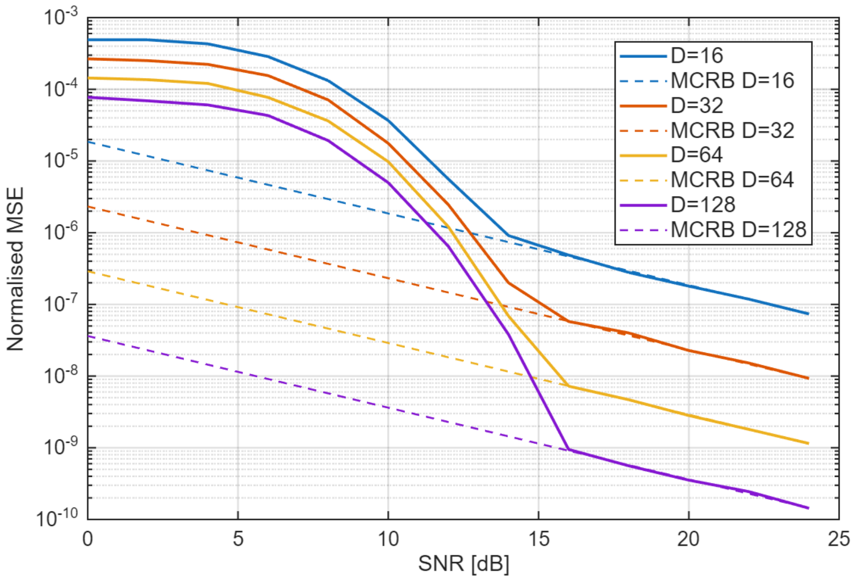


Figure 4.4: Normalised MSE of the differential phase and Kay estimator with blind operation for different observation lengths.

4.2.4 Computational Complexity

Here, the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. The real division required by the R&B parabolic interpolation is assumed to cost the same as a multiplication since it is only performed once per estimate; a look-up-table reciprocal would meet this approximation at a small accuracy cost. All analysis applies to a single polarisation. If averaging is done over both polarisations, then all operations are doubled.

Rife & Boorstyn The Rife & Boorstyn algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re\left[\frac{a+bj}{c+dj}\right] = \frac{ac+bd}{c^2+d^2}. \tag{4.9}$$

The total complexity is summarised in table 4.1; entries marked with a dagger (†) apply only to blind operation.

Operation	RM	RA
Forming $z[i], \ldots, z[i+L]$	$4L,\ 12L^\dagger$	$3L,\ 9L^\dagger$
FFT	$2L\log_2(\frac{N_{FFT}}{L}) + 2N_{FFT}\log_2(L)$	$3L\log_2(\frac{N_{FFT}}{L}) + 3N_{FFT}\log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 4.1: Computational complexity of estimating CFO with the Rife & Boorstyn algorithm.

Differential phase and Kay The differential phase and Kay estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d\, i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Differential Phase	1	$L-1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Kay	L	$L-1$

Table 4.2: Computational complexity of estimating CFO with the differential phase and Kay estimator.

4.2.5 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s. For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} for an estimation range $0 \leq |f_d/R_s| \leq 1$. In reality, a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation, then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

4.2.6 Maximum CFO

The theoretical limit on the maximum CFO that can be corrected is given by Nyquist as $0.5\,R_s$. From eq. (4.1), we see that blind operation reduces this limit by a factor of 4. Both the Rife & Boorstyn and differential phase and Kay estimators accurately estimate the CFO up to the Nyquist limit, as seen in fig. 4.5. Blind operation aliases as expected at $f/R_s = 0.125$, making it viable for CFOs up to 3.8 GHz at a symbol rate of 30.5 GBd.

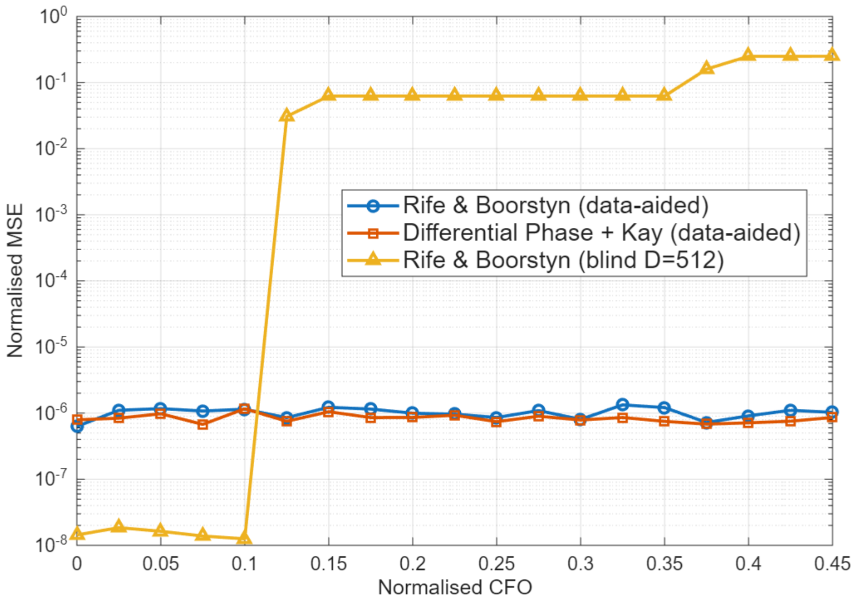


Figure 4.5: Normalised MSE in CFO estimate (MSE/R_s^2) vs. normalised CFO (f/R_s).

4.3 Phase Recovery

4.3.1 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small-angle approximations. The CPON specification sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7, the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g., 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data are used instead.

4.3.2 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of eq. (4.1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4} \arg\left(\sum_{k=1}^{N} w_{ML}^*[k]z[k]\right) - \frac{\pi}{4}. \tag{4.10}$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied to all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON specification provides one pilot symbol in every 32-symbol block that can be used to correct this [3]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{\mathrm{PU}}$ (the output of the phase unwrapper) and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{\mathrm{PU}} = \hat{\theta}_{\mathrm{PU}} - n\frac{\pi}{2}$.

4.3.3 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using eq. (4.2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm\pi/4$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

4.3.4 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 4.3.

Operation	RM	RA
Forming $z[i], \ldots, z[i+N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N+1$	$3N+2(N-1)+1$

Table 4.3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in table 4.4.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 4.4: Computational complexity of Pilots-Only correction.

4.4 Methodology

4.4.1 Phase Recovery Comparison

The two phase recovery algorithms are compared in isolation by simulating an AWGN channel with additive Wiener phase noise, without chromatic dispersion, PMD, or fibre nonlinearity, so that any performance difference is attributable solely to the phase estimation stage. BER is estimated by Monte Carlo simulation over independent realisations for each combination of laser linewidth and SNR, with the phase ambiguity inherent to

QPSK resolved by exhaustive search over the four $\pi/2$ rotations before symbol decisions are made.

4.4.2 System Energy Comparison

The full carrier recovery chain is then characterised over the same isolated channel, now with a fixed 3 GHz carrier frequency offset and Wiener phase noise at the 1 MHz linewidth limit of section 1.3.3, so the result depends only on frequency and phase recovery. Each frequency recovery algorithm is paired with each phase recovery algorithm, and for every pair the pre-FEC BER is estimated by Monte Carlo simulation over 100 independent subframes at each SNR from 0 to 30 dB. The FEC SNR is found by interpolation between adjacent SNR samples. It is computed per trial and reported as a mean with standard deviation.

Receiver energy per bit is evaluated from the per-symbol operation counts of tables 4.1 to 4.4 using the arithmetic energy model of section 2.2.1, with the frequency-recovery cost amortised over one 3712-symbol subframe and the phase-recovery cost over one 32-symbol block.

4.5 Results & Discussion

4.5.1 Phase Recovery Performance

To replace Viterbi & Viterbi with the simplified pilots-only phase correction, it must exhibit comparable performance over the range of linewidths that could be expected during operation of the network. The BER for each phase recovery algorithm as SNR varies is shown in fig. 4.6. Both algorithms have almost identical performance (plots overlapping) and are reliable up to 10 MHz, well beyond the maximum of 1 MHz given in the CPON spec.

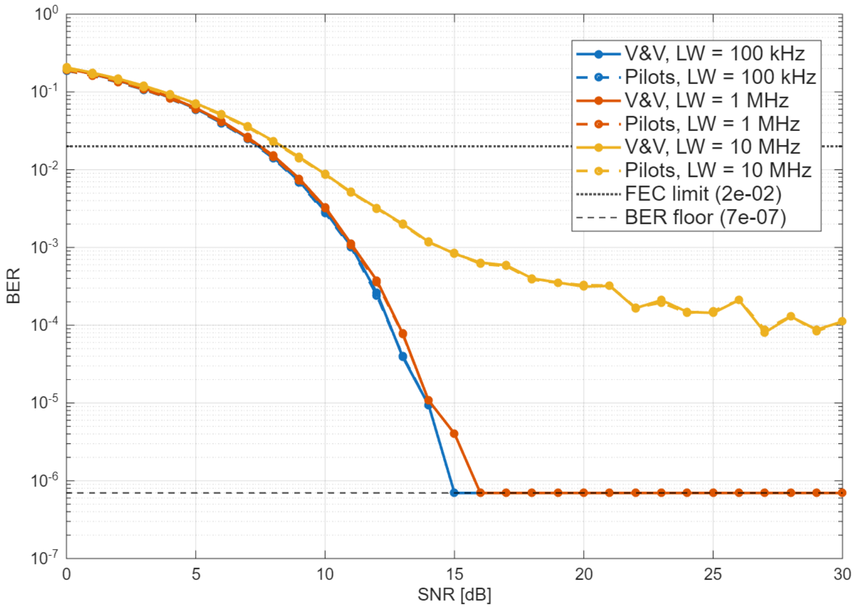


Figure 4.6: Performance comparison of V&V and Pilots-Only phase correction

The choice of fixed-point precision used in phase recovery will balance receiver energy and performance. Figure 4.7 compares the performance of both algorithms which is almost identical down to 6 bits, where the FEC SNR of Viterbi & Viterbi rises quickly. The additional operations needed by Viterbi & Viterbi to estimate the phase error mean that quantisation errors compound, degrading performance at low precision. The ability of the pilots-only estimate to match or improve on the FEC SNR of Viterbi & Viterbi is likely due to the inherent robustness of QPSK to additive and phase noise; pilots-only will produce a slightly less accurate estimate but as long as it brings the symbol within the correct QPSK quadrant, decoding will be successful.

It is then clear from fig. 4.7 that a receiver can save energy at no performance cost by using the pilots-only estimate with 4 bits of precision. The increase in transmitter energy from the slight increase in FEC SNR at 2 bits of precision may be outweighed by the receiver energy savings, but this will depend on whether the carrier recovery block limits the receiver SNR and other system parameters like the wall-plug efficiency of the laser. All further investigations in this section use pilots-only for phase recovery.

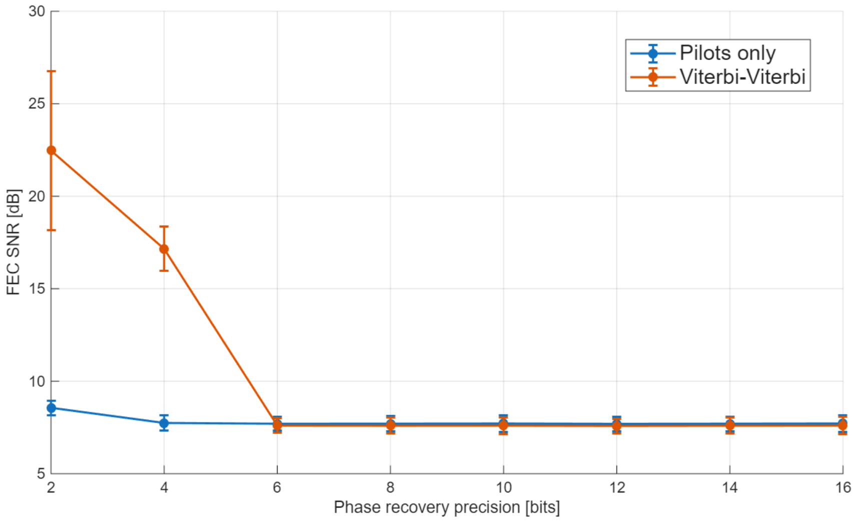


Figure 4.7: Phase recovery performance vs fixed-point precision at 1 MHz laser linewidth, with Rife & Boorstyn used to remove the initial CFO.

4.5.2 Blind vs Data-Aided Operation

The decrease in threshold SNR for longer observation lengths suggests that the performance of the Rife & Boorstyn algorithm will improve with blind observation length D. This is not the case for the differential phase and Kay estimator, where increasing blind observation length has no effect on threshold SNR. The effect of observation length on FEC SNR is shown for both algorithms in fig. 4.8, where blind Rife & Boorstyn shows a clear increase in performance over the blind differential phase and Kay estimator for longer observations and improves on data-aided Rife & Boorstyn after ≈ 50 symbols. The differential phase and Kay estimator in blind operation is thus not viable for frequency recovery, but blind Rife & Boorstyn can provide the best performance at a higher computational cost.

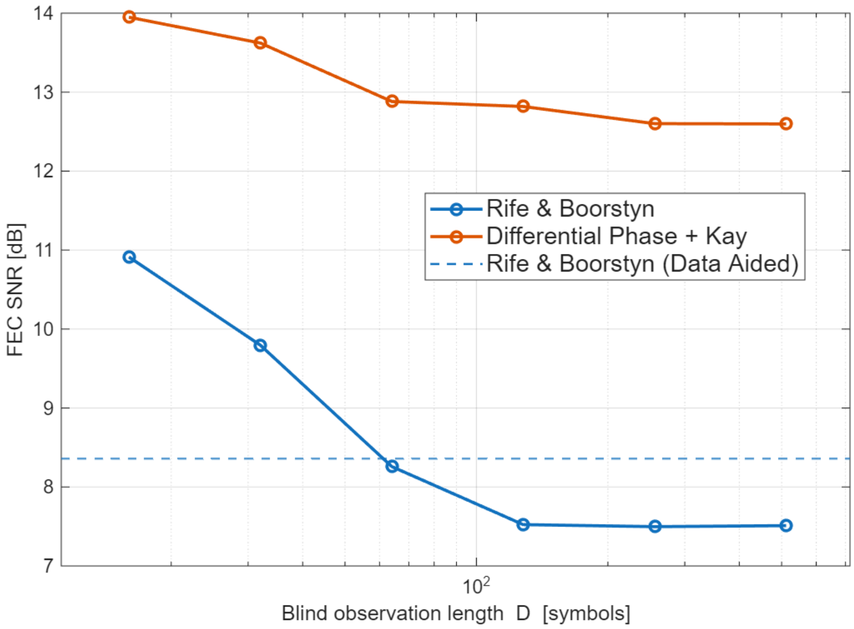
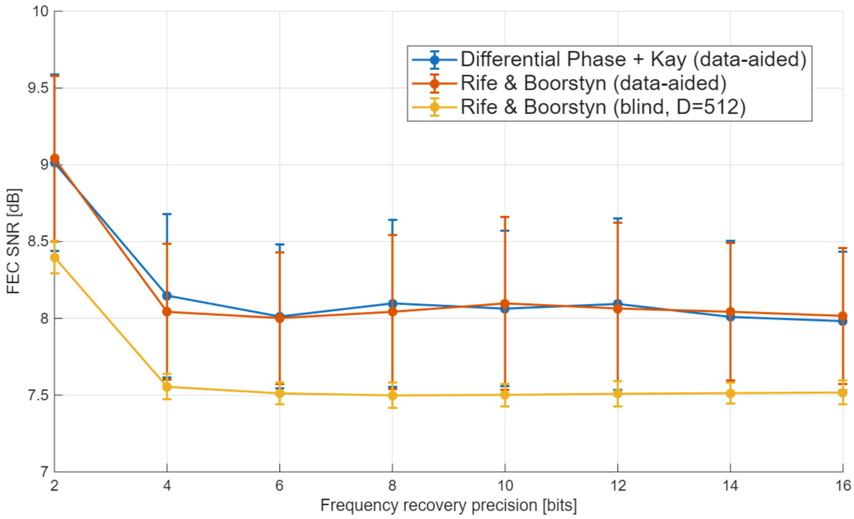


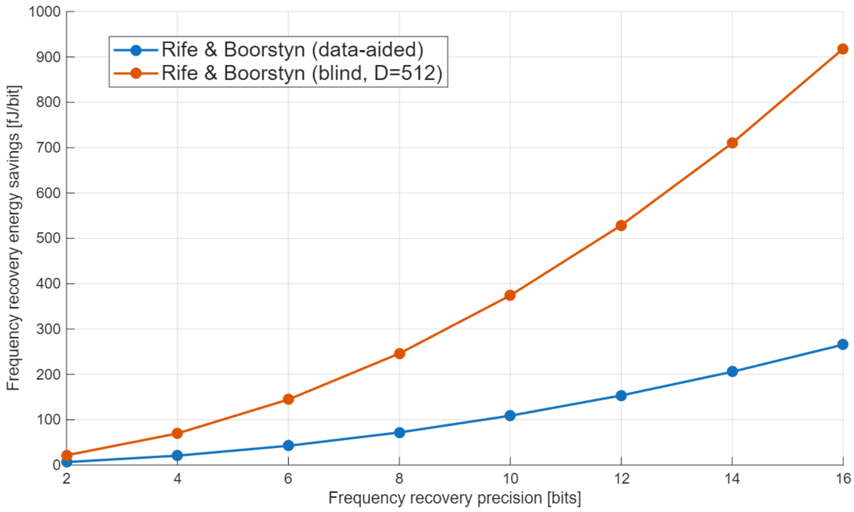
Figure 4.8: FEC SNR vs blind observation length for the Rife & Boorstyn and differential phase and Kay estimators.

4.5.3 Frequency Recovery Performance

The fixed-point performance of each frequency recovery algorithm is compared in fig. 4.9a, and the corresponding energy savings when switching to the differential phase and Kay estimator are shown in fig. 4.9b. Data-aided Rife & Boorstyn appears to offer no performance improvement on differential phase and Kay but consumes more energy. Blind Rife & Boorstyn consumes the most energy but offers a significant performance increase over both data-aided algorithms. The performance of all algorithms does not degrade until 4 bits of precision. Decreasing the precision further will again depend on the balance between transmitter and receiver energy. The most energy-efficient algorithm depends on how frequently the CFO must be estimated; frequent estimates will likely favour the differential phase and Kay estimator but in the extreme case of only one CFO estimate per transmission then amortisation makes the energy consumption negligible and blind Rife & Boorstyn will be preferred. Only using 11 symbols in the CFO estimate for data-aided operation introduces high sensitivity to the noise present in those symbols, hence a high variance in the FEC SNR is observed. This has significant design implications as a transmitter tailored to provide the average FEC SNR at the receiver could cause decoding errors if the noise on the 11 training symbols is particularly high. A safe design would then ensure the receiver SNR is closer to the maximum of the FEC SNR range. The variance could be reduced by averaging the CFO estimate over several subframes, but this requires the CFO to stay constant for long enough and consumes more energy at the receiver.



(a) FEC SNR



(b) Energy saved by switching to the differential phase and Kay estimator

Figure 4.9: Frequency recovery as fixed-point precision varies, comparing data-aided differential phase and Kay, data-aided Rife & Boorstyn, and blind Rife & Boorstyn, with the corresponding energy savings when switching to differential phase and Kay. FR energy amortised over one subframe (3712 symbols).

4.5.4 Energy Breakdown

Blind Rife & Boorstyn and differential phase and Kay with pilots-only phase recovery are the two most promising approaches for energy-efficient carrier recovery. The energy breakdowns for each are shown in fig. 4.10, and the combined energies and FEC SNRs are listed in table 4.5. Estimating the phase error with pilots-only consumes very little energy, and blind R&B dominates energy consumption in estimating CFO even with amortisation over a subframe. Differential phase and Kay with 4 bits of precision offers a lower FEC SNR for less energy than blind Rife & Boorstyn with 2 bits of precision, so the most energy-efficient implementation will select one of the other 3 combinations depending on the balance between transmitter and receiver energy.

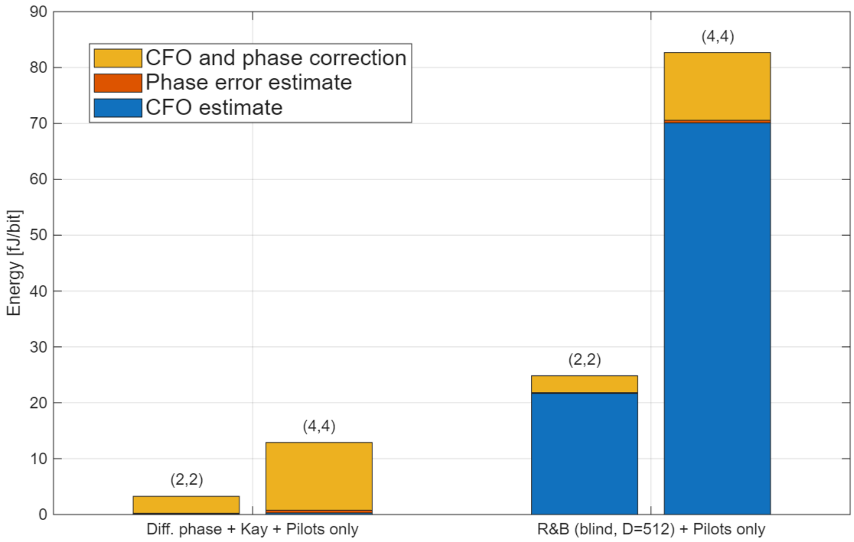


Figure 4.10: Receiver energy breakdown of carrier recovery for 4 bits (left) and 2 bits (right) of precision.

FR algorithm	Fractional bits	E_{FR} (fJ/bit)	E_{CR} (fJ/bit)	E_{apply} (fJ/bit)	E_{total} (fJ/bit)	FEC SNR (dB)
Diff. phase + Kay	2	0.08	0.14	3.03	3.26	9.76 ± 0.89
	4	0.29	0.48	12.12	12.89	8.11 ± 0.50
R&B (blind, $D = 512$)	2	21.66	0.14	3.03	24.83	8.96 ± 0.16
	4	70.08	0.48	12.12	82.68	7.60 ± 0.09

Table 4.5: Combined energy and FEC SNR of the differential phase and Kay estimator and blind Rife & Boorstyn, both using pilots-only phase recovery.

4.6 Conclusions

The robustness to both additive and phase noise of QPSK allows the phase recovery block of a coherent receiver to be dramatically simplified by just using pilot symbols with no noticeable drop in performance, resulting in considerable energy savings. Frequency recovery can also be implemented with simple algorithms to save energy; however, the efficacy of doing so depends on how often the CFO must be estimated.

Chapter 5

Full DSP Pipeline

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